

Laboratory Test - Week 9

A character string representing the name of a text file is defined in the data segment. The file contains a sequence of a maximum of 100 numbers from the interval $[-10^9, 10^9]$ that ends with the value 0 (which is not part of the array).

1. Read the content of the file and save the sequence in memory.
2. Go through the saved sequence of numbers and construct the sequence of partial sums in memory.
3. Read the name of a new file from the keyboard (from the terminal), followed by pairs of numbers of the form $a\ b$ until the pair $-1\ -1$ is encountered, and calculate the sum of the elements corresponding to the subsequence that starts at position **a** and ends at position **b**. All calculated sums will be displayed one per each line in the file read from the terminal.

Example

The text "input.txt" is defined in the data segment. The file **input.txt** has the following content:

10 3 -2 7 -5 4 -8 20 0

The file named "output.txt" is read from the keyboard, the the following pairs are read:

0 4
1 3
2 5
-1 3
5 7
3 8
-1 -1

In the file **output.txt**, the following content will be displayed on each line:

13
8
4
16

Observations

- You must validate the interval that is read from the keyboard and not display anything in the output file if the interval is not valid.
- The sequence of the partial sums of the sequence x is represented by array y , which is defined as follows:

$$y[k] = \sum_{i=0}^k x[i], \text{ for } k = 0..n-1;$$

- Thus, the subsequence that starts at position a and ends at position b has the sum $y[b] - y[a-1]$;
- Note that the indexing of each sequence starts from 0.